SOUMYADEEP ROY

Stanford University, Division of Computational Medicine, Department of Medicine | soroy@stanford.edu | https://roysoumya.github.io | https://www.linkedin.com/in/roysoumyadeep | https://github.com/roysoumya

PROFESSIONAL SUMMARY

Postdoctoral Scholar at Stanford University working at the intersection of biomedical AI, machine learning, and natural language processing. Designs and evaluates clinical AI systems using real-world patient data across Parkinson's disease, oncology, and postoperative pain management. Research spans LLM reliability evaluation, guideline-grounded clinical decision support, efficient pretraining of biological foundation models, and domain adaptation for medical text. Published at top-tier venues including ACL, EMNLP, SIGIR, IJCAI, and ECAI.

EDUCATION

Indian Institute of Technology Kharagpur — Kharagpur, India

PhD, Computer Science and Engineering. Thesis: Domain Adaptation for Medical Language Understanding. — *07/2025*

Indian Institute of Technology Kharagpur — Kharagpur, India

MS (by Research), Computer Science and Engineering. Thesis: Computational Approaches for Online Monitoring [Modeling, Analysis and Recommendation]. GPA: 9.21 — *11/2019*

Kalyani Government Engineering College — Kalyani, West Bengal, India

B.Tech, Computer Science and Engineering. Thesis: Understanding Email Interactivity and Predicting User Response to Email. GPA: 8.61. — *06/2017*

EXPERIENCE

Postdoctoral Scholar, Division of Computational Medicine, Dept. of Medicine — 09/2025 - Present

STANFORD UNIVERSITY — *United States*

- Faculty Advisor: Prof. Tina Hernandez-Boussard.
- Building and evaluating LLM-based clinical decision support systems, specifically on two applications: (i) clinical guideline adherence over real-world patient trajectories and (ii) synthetic cohort generation for evaluating AI-based patient-to-trial matching systems like TrialGPT.
- Specifically, develop graph-based hybrid LLM models to analyze guideline deviations in real-world patient trajectories (longitudinal EHR data and clinical notes)
- Guest Lecturer, BMDS 223 'Deploying and Evaluating Fair AI in Healthcare' (Spring 2026): delivered lecture on 'Bias Evaluation in LLMs' and hands-on coding workshop on 'Bias Audit on Real-World Data (MIMIC-IV)'.

PhD AI Intern, HealthCare Technology and Innovation Center — 11/2024 – 05/2025

WIPRO GE HEALTHCARE PVT. LTD — *Bangalore, India*

- Built generative AI-based dashboard for oncology patient data using Chroma vector database and LangChain, enabling clinicians to navigate complex patient histories through semantic search.
- Developed deep learning-based representation learning system for sensor log data from medical imaging equipment to enable anomaly detection and predictive maintenance; resulted in one patent filing.
- Collaborated with multi-disciplinary research teams to develop Proof-of-Concept models ready for engineering handoff.

Research Associate, L3S Research Center — 01/2021 – 06/2023

LEIBNIZ UNIVERSITY HANNOVER — *Hannover, Germany*

- Led multi-institutional research collaboration with Hannover Medical School on Parkinson's Disease patient subtyping using the Michael J. Fox Foundation LRRK2 dataset (1,500+ patients with structured EHR data).
- Developed interpretable decision tree-based patient subtyping method, identifying six novel subtypes externally validated on the Movement Disorders Society dataset; published in Frontiers in Artificial Intelligence (Impact Factor: 6.7).
- Developed pointwise mutual information-based token masking technique for DNA transformer models (DNABert, LOGO), achieving 10x speedup for few-shot gene sequence classification; published at ECAI 2023.

Research Intern, Big Data Experience Lab — 05/2018 – 06/2018

ADOBE SYSTEMS INDIA PVT. LTD — *Bangalore, India*

- Built deep learning classification model for brand personality detection from Fortune 500 company websites and a sentence-level text recommendation model for brand-aligned content rewriting; resulted in one conference paper (ACM WebSci 2019) and one journal paper (ACM Transactions on the Web, TWEB 2021).

- Designed and executed large-scale data annotation project on Amazon Mechanical Turk with quality control mechanisms.

Skills

Deep Learning: PyTorch, Transformers, Parameter-Efficient Fine-Tuning (LoRA, QLoRA)
NLP and LLMs: Fine-tuning, Prompt Engineering, RAG Systems, Model Evaluation, Efficient Training
Healthcare AI: Clinical NLP, EHR Analysis, Medical Knowledge Graphs (UMLS, PrimeKG), MIMIC-III/IV
ML Tools: Weights and Biases, Docker, Git
Programming: Python, R, SQL, Bash
Data Collection and Analysis: Potato, Amazon Mechanical Turk, Prolific

Research Interests

Biomedical NLP | Efficient Model Training | Reliability Evaluation of Clinical LLM/RAG/Agent | Clinical Decision Support | Clinical Guideline Adherence | Graphs | Translational Research (Safety Evaluation, Interpretable-by-design Models)

Publications And Research Contributions

<u>Efficient Model Training and Domain Adaptation</u>

- [ECAI 2023] Developed GeneMask, a masked language modeling masking technique to jointly mask highly correlated tokens, reducing model pretraining time and achieving 10x speedup for few-shot genomic sequence classification. doi: https://doi.org/10.3233/FAIA230492 | code: https://github.com/roysoumya/genemask
- [ECAI 2024] Designed adaptive curriculum masking strategy for DNA transformer models achieving improved performance on genome understanding benchmarks for both full-dataset and few-shot gene sequence classification. doi: https://doi.org/10.3233/FAIA240864 | code: https://github.com/roysoumya/curriculum-GeneMask
- [IJCAI 2024] Proposed vocabulary adaptation method for medical text summarization, reducing fine-tuning time from 450 days to 45 hours through fragment score-based hyperparameter optimization. doi: https://doi.org/10.24963/ijcai.2024/683 | code: https://github.com/gb-kgp/MEDVOC
- [EMNLP 2024 Findings] Developed adaptive BPE tokenization technique that modifies the initialization step to better preserve domain-specific terminology during fine-tuning. doi: https://doi.org/10.18653/v1/2024.findings-emnlp.863 | code: https://github.com/gb-kgp/adaptbpe
- [ACL 2026 Mains] 'Learning Faster with Better Tokens: Parameter-Efficient Vocabulary Adaptation for Specialized Text Summarization.' Augments pretrained tokenizers with domain-specific tokens while selectively replacing under-trained tokens; reduces training time by 35-55% over continual pretraining and reduces parameter counts by 37% (to appear, San Diego, July 2026).

<u>Medical LLM/RAG/Agent Evaluation</u>

- [Under Review] 'Cascading Tail Risk: Trajectory-Level Safety Evaluation for Sequential Clinical LLM Agents': developed trajectory-level evaluation framework with novel metrics (Tail Harm Escalation and Tail-Restricted Wasserstein Distance) for surfacing rare, compounding safety failures in sequential clinical LLM agents, demonstrated on opioid prescribing in postoperative pain management.
- [Under Review] Built Q-Pain++, a factorial stress-testing and multi-modality evaluation framework that audited five clinical LLMs and exposed a decision-space framing effect driving ~92% of treatment decision changes, showing binary safety audits are inadequate for frontier clinical models.
- [SIGIR 2024] Conducted systematic evaluation of GPT-4 on USMLE medical licensing exam questions, creating comprehensive error taxonomy identifying 8 distinct failure modes; released publicly available dataset with 250+ annotated errors. doi: https://doi.org/10.1145/3626772.3657882 | code: https://github.com/roysoumya/usmle-gpt4-error-taxonomy
- [ACL 2025 Findings] Demonstrated that high out-of-vocabulary concentration in medical texts leads to significant performance degradation in summarization tasks, with vocabulary adaptation providing effective mitigation. doi: https://doi.org/10.18653/v1/2025.findings-acl.1179 | code: https://github.com/gb-kgp/LLM-MedicalSummarization-Benchmark
- [WSDM 2025] Co-organized tutorial 'Building Trustworthy AI Models for Medicine: From Theory to Applications' covering evaluation protocols, robustness assessment, and deployment considerations. doi: https://doi.org/10.1145/3701551.3703477 | tutorial: https://sites.google.com/view/trustworthy-medical-ai/
- [LREC 2026] Created LongTailQA, a disambiguated benchmark addressing 8-30% entity ambiguity in existing long-tail QA datasets, revealing RAG models gain up to 24.7% accuracy improvement from query disambiguation. doi: http://www.lrec-conf.org/proceedings/lrec2026/pdf/2026.lrec2026-1.405.pdf | code:

https://github.com/williamx854/LongTailQA-Benchmark | data: https://huggingface.co/datasets/willx7890/LongTailQA

AI Development and Evaluation with Real-World Patient Data

- [Under Review] Submitted decision-aware evaluation suite and novel scale-independent metric (Uncertainty Index), showing standard uncertainty quantification methods systematically fail to flag high-risk errors in clinical ML across three real-world datasets.
- [Frontiers in AI 2025] Published decision tree-based approach for Parkinson's Disease patient subtyping using clinical and genomic data, validated by medical experts. doi: https://doi.org/10.3389/frai.2025.1668206 | code: https://github.com/roysoumya/decision-tree-subtyping-Parkinson

Clinical NLP Applications

- [CIKM 2021] Built knowledge-aware dual-encoder neural network for medical forum question classification, prioritizing medical concept-bearing words extracted using the QuickUMLS tool. doi: https://dl.acm.org/doi/10.1145/3459637.3482128 | code: https://github.com/roysoumya/knowledge-aware-med-classification
- [ICDH 2023] Developed interpretable clinical trial search system using PubMed citation network to improve retrieval via metapath-based similarity search and improved aspect-based ranking fusion (Best Student Paper Candidate). doi: https://doi.org/10.1109/ICDH60066.2023.00056 | code: https://github.com/roysoumya/MPSS-clinical-trial-search
- [ECAI 2025] Mentored a PhD student through a multi-institution collaboration, creating annotated dataset for physician intent modeling in doctor-patient dialogues and demonstrating that intent filtering improves dialogue summarization performance. doi: https://doi.org/10.3233/FAIA251359 | code: https://github.com/DATEXIS/medical-intent-classification

Applied And Translational Work

- [Stanford Postdoc Project, 2025-present] Developing computable guideline pathways and computing deviations in patient care trajectories using Stanford Healthcare and MIMIC data (structured EHR and clinical notes).
- [Boston GrandHack 2026] Co-developed AuriCare, a holistic pain management decision support concept, presented at MIT Hacking Medicine's Boston GrandHack 2026. blog: https://medium.com/@soumyadeeproy/building-auricare-in-48-hours-an-ai-pain-calibration-system-at-mit-grandhack-2026-fdb99e2683ab | event: https://hackingmedicine.mit.edu/events/grandhack-2026
- [Patent, filed 2025] Co-inventor on patent filing for deep learning-based representation learning system for medical imaging equipment sensor logs, enabling anomaly detection and predictive maintenance (Wipro GE Healthcare).

Professional Activities

- Memberships: Sigma Xi, The Scientific Research Honor Society; Association for Computational Linguistics; AMIA Digital Member.
- Reviewing (CORE A* CS Conferences): EMNLP 2022, ACL 2023, EMNLP 2023, AAAI 2023-2026, ACL Rolling Review February/May/July 2025, January/March 2026, FAccT 2026.
- Reviewing (CORE A CS Conferences): COLING 2025, ML4H 2024, EACL 2023, ECML-PKDD 2020.
- Reviewing (Medical AI and Informatics): AMIA 2026, Journal of the American Medical Informatics Association, Frontiers in AI, Knowledge and Information Systems, Frontiers in Genetics, IEEE Transactions on Cognitive and Developmental Systems, Elsevier Artificial Intelligence in Medicine.

Tutorials And Invited Talks

- January 2026: Invited talk, 'Foundation Models for Medical Text,' AI Foundation Models in Biomedicine course, Leibniz University Hannover, Germany. Slides: https://docs.google.com/presentation/d/1fi_lvu36drjCtUuK0zg8gHBIj723clTr_Gqn0aa_7MA/edit?usp=sharing
- January 2026: Invited talk, 'Building a Career in AI for Industry and Academia: Current and Future Trends,' Google Developers Group on Campus, Kalyani Government Engineering College, India. Website: https://gdg.community.dev/events/details/google-gdg-on-campus-kalyani-government-engineering-college-kalyani-india-presents-building-a-career-in-ai-for-industry-and-academia-current-and-future-trends/ | Slides: https://docs.google.com/presentation/d/1bDDa2VyuCVMuOoc07Z-M4-E99oBf-qGSf0fs54b_N_A/edit?usp=sharing
- December 2025: Invited talk, 'Vocabulary Adaptation Strategies for Finetuning Medical Language Models: Towards Reliable Medical Summarization,' Breakfast Talk series, Microsoft Research India, Bangalore. Slides: https://docs.google.com/presentation/d/1NwEj-9VXE86W_k5NXepmew21UZu0R4ulI4UEcLWfYhk/edit?usp=sharing
- August 2025: Invited presentation of ECAI 2023 and 2024 work on 'Improving Training Efficiency of DNA Foundation Models,' 1st Symposium on Intersections: Statistical Genomics and Complex/Chronic Disease Biology, BRICS - National Institute of Biomedical Genomics, Kalyani, West Bengal.

- March 2025: Conference tutorial, 'Building Trustworthy AI Models for Medicine: From Theory to Applications,' 18th ACM International Conference on Web Search and Data Mining (WSDM 2025). Tutorial website: https://sites.google.com/view/trustworthy-medical-ai/ | Slides: https://docs.google.com/presentation/d/1wWaxOfIdaFc59t21EHdpQhqW7ZLlD9QVZjaQjEFd1jg/edit?usp=sharing
- December 2024: Invited talk, 'Foundation Models for Medical Text,' AI Foundation Models in Biomedicine course, Leibniz University Hannover, Germany. Slides: https://docs.google.com/presentation/d/1vdJ1dkhz17QvvgmOe2il8KCCN8gLMdi58xddzcrNrjI/edit?usp=sharing

TEACHING

- Stanford University: Guest Lecturer, BMDS 223 'Deploying and Evaluating Fair AI in Healthcare' (Spring 2026). Course: https://biomedin223.su.domains/2026/ | May 5 Lecture: https://roysoumya.github.io/assets/pdf/BMDS223_May5_Lecture_Soumyadeep.pdf | May 7 Coding Workshop: https://roysoumya.github.io/assets/jupyter/workshop_bias_audit_MIMICIV_May7.ipynb
- IIT Kharagpur: Programming and Data Structure Lab (Spring 2018), Department Website Team (Autumn 2018-Spring 2020), Information Retrieval (Autumn 2020), Computing Lab 1 (Autumn 2023).
- Leibniz University Hannover, Germany: Natural Language Processing (Winter 2021), Foundations of Information Retrieval (Summer 2022, 2023).